AMPLENESS VERSUS VERY AMPLENESS OF POWERS FOR FINITE TYPE MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-morphisms/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot. Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

Editorial note. The citation fields (source_id, source_version, authors, source_url, license.name, source_title) are identical to those of every other card this lane has produced from this chapter, so the batch selector treats them as one source.

Editorial note. Read in full under the Lead's 12:4x rule. The proof contains 'Trivially (6) implies (5)' / 'Trivially (4) implies (3)' - the word marks genuinely trivial implications between the listed conditions, and every non-trivial step is a cited result, so the assembly of the equivalence is the author's own work.

Allowed standard prerequisites (editorial).

- Very ampleness implies ampleness for quasi-compact morphisms. Lemma \ref{lemma-ample-very-ample}, cited at proof lines 9566-9568. Gives (2) implies (1).
- The affine characterisation of ampleness versus very ampleness of powers. Lemma \ref{lemma-finite-type-over-affine-ample-very-ample}, cited at proof lines 9570-9571. Gives the statement on each member of an affine cover.
- Relative very ampleness is local on the base. Lemma \ref{lemma-relatively-very-ample}, cited at proof lines 9572-9573. Assembles the local statements.

Source reference. Original lemma 01VU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2459–2459. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 01VN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2452–2452. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 01VT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2458–2458. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01VR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2456–2456. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $f : X \to S$ be a morphism of schemes. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. Assume S quasi-compact and f of finite type. The following are equivalent*

(1) *$\mathcal{L}$ is f-ample,*
(2) *$\mathcal{L}^{\otimes d}$ is f-very ample for some $d \geq 1$,*
(3) *$\mathcal{L}^{\otimes d}$ is f-very ample for all $d \gg 1$.*

Proof. Trivially (3) implies (2). Lemma 01VN guarantees that (2) implies (1) since a morphism of finite type is quasi-compact by definition. Assume that $\mathcal{L}$ is f-ample. Choose a finite affine open covering $S = V_1 \cup \ldots \cup V_m$. Write $X_i = f^{-1}(V_i)$. By Lemma 01VT above we see there exists a d_0 such that $\mathcal{L}^{\otimes d}$ is relatively very ample on X_i/V_i for all $d \geq d_0$. Hence we conclude (1) implies (3) by Lemma 01VR. □